

Comparative outcomes and toxicities for ruthenium-106 versus palladium-103 in the treatment of choroidal melanoma

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For the treatment of choroidal melanoma, palladium-103 (¹⁰³Pd) and ruthenium-106 (¹⁰⁶Ru) plaque brachytherapy shows reduced toxicity compared with the historical standard iodine-125. No report has directly compared the clinical outcomes between ¹⁰³Pd and ¹⁰⁶Ru, and the reasons for the selection of one over the other remain purely theoretical. Patients with choroidal melanoma with apical tumor height up to 5 mm were included. Patients from Emory University were treated with ¹⁰³Pd between 1993 and 2012. Patients from Cleveland Clinic were treated with ¹⁰⁶Ru between 2005 and 2010. Medical records were retrospectively reviewed. We compared post-treatment visual acuity (VA), toxicity, and oncologic outcomes. ¹⁰³Pd patients ($n = 124$) and ¹⁰⁶Ru patients ($n = 42$) had a median follow-up of 4.2 and 5.0 years, respectively. Radiation retinopathy-free survival was similar for both radioisotopes, but ¹⁰⁶Ru had lower grades of retinopathy ($P = 0.006$). ¹⁰³Pd was associated with worse VA preservation ($\geq 20/40$) by year 3 (odds ratio: 3.8; 95% confidence interval: 1.01–14.31, $P = 0.048$). ¹⁰³Pd was associated with higher distant metastases-free survival (DMFS) in multivariate analysis (hazard ratio: 0.10; 95% confidence interval: 0.02–0.38; $P < 0.001$). ¹⁰⁶Ru had lower grades of radiation retinopathy and improved long-term VA preservation, but also inferior

DMFS, compared with ¹⁰³Pd. Because of the inherent limitations of a retrospective analysis, the significance of the inferior DMFS for ¹⁰⁶Ru remains unclear, although the suggestion of a slight inferiority in terms of DMFS for ¹⁰⁶Ru is consistent with the other limited literature. On the basis of this study, we believe that both radioisotopes remain appropriate for the treatment of small choroidal melanomas up to 5 mm in apical height. *Melanoma Res* 28:120–125 Copyright © 2018 Wolters Kluwer Health, Inc. All rights reserved.

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Introduction

This is a retrospective study that compares the outcomes for patients treated for choroidal melanoma with the plaque brachytherapy radioisotopes ruthenium-106 (¹⁰⁶Ru) or palladium-103 (¹⁰³Pd). ¹⁰⁶Ru patients were treated at Cleveland Clinic and ¹⁰³Pd patients were treated at Emory University. ¹⁰⁶Ru showed lower grades of radiation retinopathy and improved 3-year visual acuity (VA) preservation – but also lower distant metastasis-free survival – compared with ¹⁰³Pd.

Ocular melanoma is the most common primary intraocular tumor in adults and the second most common type of melanoma after cutaneous, accounting for 3.7% of all cases. It arises from melanocytes, usually in the uvea

(82.5%), or less frequently in the conjunctiva. Within the uveal tract, 86.3% occur in the choroid; iris and ciliary body melanoma are much less common. The incidence of choroid melanoma in the USA is ~4.3 cases per million persons; melanoma of the iris and ciliary body contributes another 0.7 cases per million persons. Patients are typically diagnosed between the ages of 50 and 80 years, with peak incidence in the seventh decade [1].

Plaque radiation therapy (RT), with proper tumor selection, leads to local control and survival outcomes similar to those achieved by enucleation, as reported in the Collaborative Ocular Melanoma Study (COMS). In this trial, which enrolled from 1986 to 2003, patients with medium-sized tumors were randomized to enucleation versus iodine-125 (¹²⁵I) plaque brachytherapy. Long-term analysis showed no difference in overall survival or distant metastases [2–4]. However, although 70% of

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patients had baseline VA 20/40 or better, only 34% of patients showed 20/40 vision or better at 3 years [5]. Some of the factors that influence post-treatment VA include tumor characteristics (mainly size and distance from the fovea), age, and ocular comorbidities such as diabetes. Secondary effects of radiation on normal ocular structures – namely radiation retinopathy (ischemic, exudative, hemorrhagic, or atrophic), radiation choroidopathy, radiation optic neuropathy, and radiation-induced cataract – are also implicated as contributors to loss in VA [6].

The most commonly used radioisotopes for ocular melanoma brachytherapy are ¹⁰⁶Ru, ¹⁰³Pd, and ¹²⁵I (please see Table 1 for a summary of the characteristics of these radioisotopes). Because of the relatively gradual dose fall-off, ¹²⁵I delivers a higher dose to critical eye structures including the macula, lens, and optic discs. Finger *et al.* [7–9] report that compared with ¹²⁵I, ¹⁰³Pd offers an opportunity for radiation sparing of ocular structures while maintaining better VA and equivalent rates of local control. ¹⁰³Pd, because of its lower energy resulting in steeper isodose fall-off, has been shown to decrease dose to the macula and optic nerve relative to the ¹²⁵I [9]. We recently reported an analysis of patients treated with plaque brachytherapy at Emory with either ¹⁰³Pd versus ¹²⁵I, which showed that ¹⁰³Pd is associated independently with superior long-term VA preservation and lower incidence of radiation retinopathy compared with ¹²⁵I [10]. ¹⁰⁶Ru, a β emitter rather than a γ emitter, has an even sharper dose fall-off than even ¹⁰³Pd. Dosimetric comparison of ¹⁰⁶Ru and ¹²⁵I for tumors less than or equal to 5 mm shows translation of this sharp dose fall-off to a clinically relevant reduction in radiation dose to normal eye structures [11]. In addition, a recent retrospective report from MD Anderson showed equivalent overall survival, improved enucleation-free survival (EFS), and reduced radiation retinopathy in patients receiving ¹⁰⁶Ru compared with patients receiving ¹²⁵I plaque brachytherapy [12].

To date, there are no comparative reports of ¹⁰⁶Ru and ¹⁰³Pd brachytherapy for choroidal melanoma. Such a report may provide further insight into outcome

implications of plaque brachytherapy radioisotope selection, particularly with respect to quality of life, vision preservation, and toxicities. Therefore, in this multi-institutional study, we compare the VA, toxicity, and oncologic outcomes for patients with choroidal melanoma treated with ¹⁰³Pd versus ¹⁰⁶Ru plaque brachytherapy.

Patients and methods

Patient selection

We obtained approval from the institutional review boards of Cleveland Clinic and Emory University for this study. We reviewed the medical charts of all patients treated for choroidal melanoma with definitive intent with ¹⁰³Pd plaque brachytherapy at Emory University between 1993 and 2012 and ¹⁰⁶Ru plaque brachytherapy at Cleveland Clinic between 2005 and 2010. Inclusion criteria were apical tumor height up to 5 mm and age older than 18 years. Tumors with an apical height of more than 5 mm are not normally treated with ¹⁰⁶Ru as the sharp dose fall-off of this isotope would necessitate a high dose to the sclera to achieve a proper dose distribution to the tumor in the choroid and such a dose to the sclera could result in scleral necrosis. We excluded patients with incomplete demographic data, pretreatment VA data, RT treatment data, tumor located in the iris or ciliary body, and metastatic disease at initial presentation. We also excluded patients who had undergone transpupillary thermotherapy.

Patients were treated at both institutions with a prescribed RT dose of 85 Gy as per the COMS trial [13]. We have previously published the detailed treatment protocols for Emory University and Cleveland Clinic [10,14]. Briefly, plaque size was chosen on the basis of tumor size, allowing for a 2 mm margin around the circumference of the tumor. For tumors more than 5 mm in height, RT dose was prescribed to a height of 5 mm.

Medical record review

Baseline patient data included age at the time of treatment, sex, presence or absence of diabetes mellitus, best-corrected VA before treatment, and tumor characteristics including laterality, location (o'clock), largest radial dimension, area, and height. Treatment data included date of plaque placement, radioisotope (¹⁰⁶Ru or ¹⁰³Pd), plaque size, RT dose, and distance of the tumor to the optic nerve (measured from the edge of the tumor to the center of the disc) and macula (measured from the edge of the tumor to the fovea). We also determined the grade of radiation retinopathy, if any, per the Finger classification [15]. Follow-up data included length of follow-up, VA at annual follow-up visits, date of diagnosis of complications of radiotherapy (e.g. radiation retinopathy), development of distant metastases, secondary enucleation, and date of death from all causes of mortality. Baseline and follow-up VA measurements were recorded using a Snellen VA scale. Count fingers VA was converted into a Snellen VA

Table 1 Isotope characteristics

	¹²⁵ I	¹⁰³ Pd	¹⁰⁶ Ru
Energy (keV)	35.5	21	39.4
Radiation type	γ	γ	β
Half-life (days)	60	17	374
Cost	\$	\$\$	\$\$\$\$
Advantages	Easy to shield Treats thicker tumors	High dose rate	Sharp dose fall-off Reusable (longer implant duration over time)
Disadvantages	Higher radiation to normal tissues	Short half-life	Only thin tumors <5 mm

¹²⁵I, iodine-125; ¹⁰³Pd, palladium-103; ¹⁰⁶Ru, ruthenium-106.

recording value [16]. Hand motion only and light perception only vision were recorded as a decimal acuity of 0.005, and no light perception and enucleation were assigned a decimal acuity of one line worse than the lowest decimal VA (0.0015) [17].

Statistical analysis

We analyzed three VA measures to determine the toxicity of the radioisotopes: visual preservation, loss, and change. VA preservation was defined as post-treatment VA of 20/40 or better. VA loss was defined impairment in vision to 20/200 or worse. Degree of change was categorized into stable/improved [e.g. VA change of $\log\text{MAR}(0.1) \geq 0$] or a decrease in lines [e.g. VA change of $\log\text{MAR}(0.1) < 0$].

Survival was defined as time from the date of plaque placement to one of the following: death [overall survival (OS)], distant metastases-free survival (DMFS), EFS, radiation retinopathy free survival (RRFS), censored at last follow-up. Survival was estimated using the Kaplan–Meier method, and survival distributions were compared using log-rank tests. Firth's penalized maximum likelihood estimation was implemented for the DMFS and EFS models to reduce bias in the parameter estimates and confidence intervals because of the small number of events in each model. Treatment groups and vision endpoints were compared across categorical endpoints using χ^2 -tests or Fisher's exact tests, where appropriate, and across continuous variables using analysis of variance. Multivariable analysis (MVA) for time-to-event endpoints was carried out using Cox proportional hazards modeling and included the following factors for OS, DMFS, and EFS: age at the time of the procedure, tumor area, tumor height, radioisotope. MVA for RRFS and VA endpoints (loss, preservation and change at 1, 2, and 3 years) included: radioisotope, pretreatment vision, distance from the macula to the tumor, distance from the optic disk to the tumor, tumor area, tumor height, and age at the time of the procedure. MVA for VA endpoints at each time point was carried out using logistic regression modeling.

All analyses were carried out using the SAS 9.3 version statistical software package (SAS Institute, Cary, North Carolina, USA). All statistical analyses were two-sided, and *P* values less than 0.05 were considered statistically significant.

Results

At Emory University, 124 patients were treated with ^{103}Pd plaque brachytherapy between 1993 and 2012, with a median follow-up of 4.2 years. At Cleveland Clinic, 42 patients were treated with ^{106}Ru plaque brachytherapy between 2005 and 2010, with a median follow-up of 5.0 years. Table 2 summarizes the patient, disease, and treatment characteristics of each cohort. There were no significant differences in the cohorts except for a minor difference in prescribed dose, and also

Table 2 Baseline patient characteristics

	^{103}Pd (<i>N</i> = 124)	^{106}Ru (<i>N</i> = 42)	<i>P</i> value
Sex [<i>n</i> (%)]			
Male	59 (47.58)	16 (38.1)	0.286
Female	65 (52.42)	26 (61.9)	
Age at the time of the procedure			
Median	64.5	69.9	0.42
Diabetes			
No	104 (85.95)	34 (80.95)	0.439
Yes	17 (14.05)	8 (19.05)	
Pretreatment vision (20/x)			
≤ 20	30 (24.19)	9 (21.43)	0.334
> 20, ≤ 50	60 (48.39)	22 (52.38)	
> 50, ≤ 100	26 (20.97)	6 (14.29)	
> 100, ≤ 200	5 (4.03)	1 (2.38)	
> 200	3 (2.42)	4 (9.52)	
Laterality			
Left	58 (46.77)	18 (42.86)	0.66
Right	66 (53.23)	24 (57.14)	
Location clock hour			
12–3	32 (25.81)	16 (38.1)	0.424
3:30–6	24 (19.35)	5 (11.9)	
6:30–9	44 (35.48)	14 (33.33)	
9:30–11:30	24 (19.35)	7 (16.67)	
Largest dimension [median (range)] (mm)	10.5 (0.1–32.8)	9.7 (4.5–16.5)	0.517
Tumor area [median (range)] (mm ²)	99.5 (0.01–1075.8)	65.9 (15.9–236.9)	0.121
Height of tumor [median (range)] (mm)	2.5 (81.3–88.1)	2.5 (71.8–92.8)	0.839
Plaque size [median (range)] (mm)	18 (14–24)	15.9 (15.9–20.4)	0.131
Dose [median (range)] (Gy)	84.5 (81.3–88.1)	85.1 (71.8–92.8)	0.003
Distance from the macula to the tumor [median (range)] (mm)	4.16 (0–21.1)	4.5 (0–18)	0.296
Distance from the optic disk to the tumor [median (range)] (mm)	4.39 (0–17.4)	4.75 (1–15)	0.129
Radiation retinopathy grade (Finger classification) [<i>n</i> (%)]			
1	16 (28.57)	12 (80)	0.006
2	13 (23.21)	0 (0)	
3	20 (35.71)	2 (13.33)	
4	6 (10.71)	1 (6.67)	
No records	1 (1.79)	0 (0)	

^{103}Pd , palladium-103; ^{106}Ru , ruthenium-106.

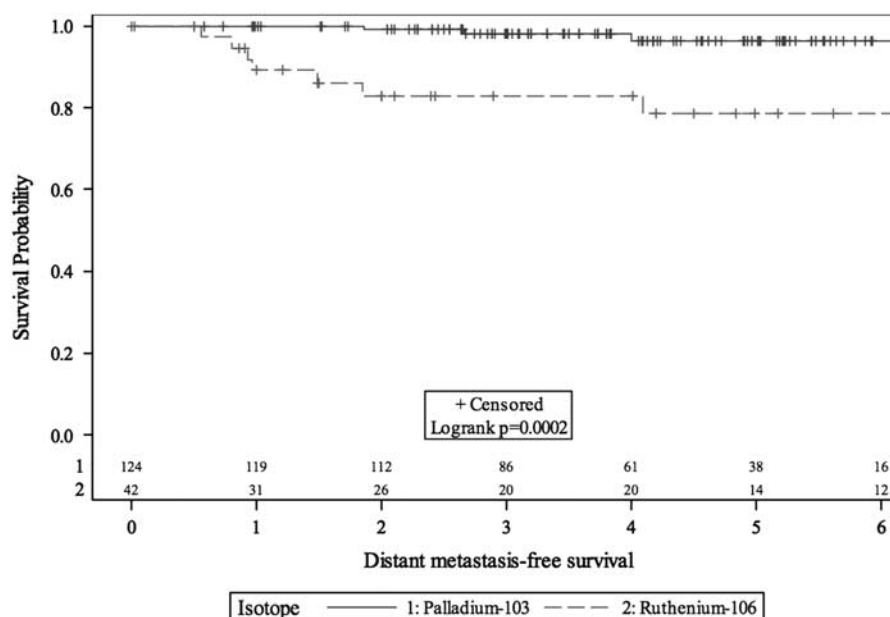
Bold values indicate significant *P* value. *P* value is calculated by analysis of variance for numerical covariates; and χ^2 -test or Fisher's exact for categorical covariates, where appropriate.

difference in radiation retinopathy grade per the Finger classification [15]. ^{106}Ru patients had lower grades of retinopathy compared with the ^{103}Pd cohort at the latest follow-up visit (*P* = 0.006). At baseline, ^{106}Ru and ^{103}Pd patients had similar distributions of pretreatment VA.

EFS could not be analyzed as there were too few events in the cohorts: two in the ^{106}Ru cohort and one in the ^{103}Pd cohort. Five-year DMFS was higher in the ^{103}Pd cohort: 96.5 versus 78.6% in the ^{106}Ru cohort (*P* = 0.0002) (Fig. 1). Five-year OS was also higher in the ^{103}Pd cohort: 89.3 versus 80.2% in the ^{106}Ru cohort (*P* = 0.0347).

Fifteen (35.7%) patients treated with ^{106}Ru and 56 (45.1%) patients treated with ^{103}Pd developed radiation

Fig. 1



Distant metastasis-free survival for ruthenium-106 versus palladium-103.

retinopathy. Three-year RRFS was not statistically different between ¹⁰³Pd and ¹⁰⁶Ru (65.2 vs. 50.6%, respectively, $P=0.5420$) (Fig. 2). Table 3 summarizes the rates of VA loss, preservation, and stability/improvement over 3 years for ¹⁰⁶Ru and ¹⁰³Pd.

The significantly higher DMFS for ¹⁰³Pd persisted in MVA [hazard ratio (HR): 0.10; 95% confidence interval (CI): 0.02–0.38; $P<0.001$]. Tumor height was associated significantly with worse DMFS in MVA (HR: 3.07; 95% CI: 1.60–5.90; $P<0.001$); no other covariates were associated significantly with DMFS. The higher OS found for ¹⁰³Pd also persisted on MVA (HR: 0.19; 95% CI: 0.07–0.54; $P=0.002$). Increasing tumor height (HR: 2.43; 95% CI: 1.38–4.27; $P=0.002$) and older age at the time of procedure (HR: 1.15; 95% CI: 1.07–1.23; $P<0.001$) were associated significantly with lower OS on MVA; other covariates were not associated significantly with OS. In MVA of rates of RRFS, increasing distance from the optic disc to the tumor was shown to have significantly higher RRFS (HR: 0.83; 95% CI: 0.74–0.93; $P<0.001$); no other covariates (including radioisotope used) were found to be associated significantly with RRFS in MVA. In MVA for VA preservation (post-treatment vision 20/40 or better), the ¹⁰³Pd radioisotope was associated with worse outcomes (odds ratio: 3.8; 95% CI: 1.01–14.31; $P=0.048$); no other covariates had significant associations with VA preservation. In MVA for VA loss (impairment in vision to 20/200 or worse), increasing distance from the optic disk to the tumor was associated with improved outcomes (odds ratio: 0.78; 95% CI: 0.63–0.97; $P=0.023$); no other covariates had significant associations with VA loss. No

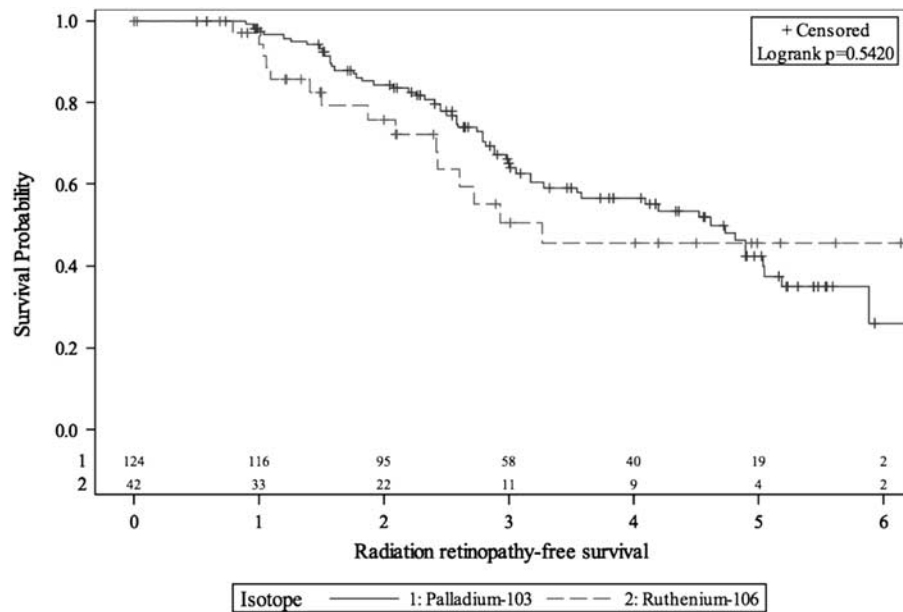
covariates (including radioisotope used) were associated significantly with VA stability/improvement in MVA.

Discussion

Since the initiation of the COMS study in 1986, there has been a sharp and steady increase in the use of radiation and decline in surgery for uveal melanoma. Surgery alone was used in 93.8% of cases for 1973–1975 compared with 28.3% for 2006–2008, with a corresponding increase in those treated with radiation from 1.8% of cases for 1973–1975 compared with 62.5% for 2006–2008 [18]. This paradigm shift has likely improved patient quality of life through ocular preservation, while maintaining DMFS and OS. In the present analysis, we sought to compare, for the first time, the oncologic, toxicity, and VA outcomes of ¹⁰⁶Ru and ¹⁰³Pd episcleral plaque brachytherapy in the definitive treatment of choroidal melanoma with apical height up to 5 mm. In this multi-institutional analysis, we found that the ¹⁰³Pd group had improved DMFS at 5 years (96.5 vs. 78.6%, $P=0.0002$). Three-year RRFS was equivalent in both cohorts, but ¹⁰⁶Ru had lower grades of retinopathy. 3-year VA preservation was worse in the ¹⁰³Pd cohort in MVA. We report improved OS with ¹⁰³Pd; however, we believe that this significant difference is confounded by the competing comorbidities and selection bias inherent to a retrospective analysis of all-cause mortality.

Although a small number of patients (4%) present with distant metastases at diagnosis, about half of the patients develop metastases in their disease course [19], which is significant because the prognosis of metastatic disease

Fig. 2



Radiation retinopathy-free survival for ruthenium-106 versus palladium-103.

Table 3 Visual acuity outcomes for ruthenium-106 versus palladium-103

Years	¹⁰⁶ Ru [n (%)]				¹⁰³ Pd [n (%)]			
	Preservation (better than 20/40)	Loss (worse than 20/200)	Stable/improved	n	Preservation (better than 20/40)	Loss (worse than 20/200)	Stable/improved	n
0	28 (67)	5 (12)	–	42	83 (67)	6 (5)	–	124
1	16 (50)	2 (6)	18 (56)	32	81 (66)	12 (10)	70 (57)	122
2	15 (68)	0 (0)	15 (68)	22	59 (56)	15 (14)	53 (50)	106
3	10 (67)	1 (7)	9 (60)	15	45 (48)	17 (18)	37 (40)	93

¹⁰³Pd, palladium-103; ¹⁰⁶Ru, ruthenium-106.

has been historically very poor. At 5 and 10 years of follow-up, choroidal melanoma metastases have been reported in 15 and 25% of patients, respectively [20]. Uveal melanoma has a high propensity for the liver (94%), followed by lung (24%) and bone (16%) [21]. Memorial Sloan Kettering reviewed their experience with ¹⁰⁶Ru for definitive management of uveal melanoma and found a 5-year DMFS of 88.6% [22]. Their cohort had a median tumor diameter of 9.4 mm and height of 2.6 mm, which is similar to Cleveland Clinic’s cohort, which had median values for tumor diameter of 9.75 mm and height of 2.5 mm. Finger *et al.* [7] reported 5-year DMFS of 92.7% for their ¹⁰³Pd cohort consisting of mainly medium-sized choroidal melanoma, which is similar to our Emory experience. Because of the inherent limitations of a retrospective analysis, the significance of the inferior DMFS found for ¹⁰⁶Ru in our study remains unclear, although the suggestion of a slight inferiority in terms of DMFS for ¹⁰⁶Ru is consistent with the other limited literature.

Multiple reports from throughout the world have detailed the VA and toxicity outcomes of ¹⁰⁶Ru, although it is somewhat difficult to compare these studies directly as each report contains unique baseline patient and tumor characteristics. In our ¹⁰⁶Ru cohort, 67% had VA 20/40 or better and 93% VA 20/200 or better at a median of 5.0 years of follow-up after treatment. Marconi *et al.* [23] reported, in their cohort of 83 ¹⁰⁶Ru patients (median tumor height 4.3 mm and diameter 9.3 mm) with a 3.25-year median follow-up, that 56% had VA 20/200 or better and 30% had VA 20/40 or better. A Swedish study reported results of 579 eyes treated with ¹⁰⁶Ru. In their patients, the median tumor height was 4.3 mm, the median diameter was 10 mm, the median distance from the macula was 2.0 mm, and the median distance from the optic disc was 3 mm. These tumors were thicker and closer to the macula and optic disc than in our 106Ru cohort. The reported 5-year VA 20/40 or better was 31% and VA 20/200 or better was 49% [24]. A report of 458 patients in the UK with choroidal melanoma (median tumor height 3.2 mm and median diameter

10.6 mm) reported that 80% had VA 20/40 or better and 93% had VA 20/200 or better at 2 years. In terms of toxicity, MD Anderson has reported that 50% of their ¹⁰⁶Ru patients have radiation retinopathy (median tumor height 3.05 mm and median diameter 9.6 mm) at median 67 months of follow-up [12]. At Helsinki University, 28% of ¹⁰⁶Ru patients (median tumor height 1.9 mm and median diameter 7 mm) experienced radiation maculopathy at 5 years [25].

As with any retrospective study, there are several limitations in the present analysis. There is potential for selection bias because of the nonrandomized treatment cohorts. However, both Emory University and Cleveland Clinic have uniform follow-up patterns, and most patients have detailed clinical records that have been maintained meticulously. Patient, tumor, and treatment characteristics were well documented at both institutions, and were well balanced between institutional cohorts. The quality of documentations made it possible to report grade of radiation retinopathy, a variable that is not found frequently in the literature. Once again, there may be confounding because of differences in surgeon experience between Emory University and Cleveland Clinic. Finally, we have relatively small cohorts, especially for ¹⁰⁶Ru brachytherapy, which is attributable to the limited availability of ¹⁰⁶Ru in the USA during the last few decades.

Conclusion

We compared oncologic, toxicity, VA acuity outcomes of patients treated with episcleral plaque brachytherapy for choroidal melanoma up to 5 mm in height, and found that ¹⁰⁶Ru yielded similar rates, but lower grades of radiation retinopathy, and superior VA preservation at 3 years, compared with ¹⁰³Pd. This advantage for ¹⁰⁶Ru in terms of toxicity is perhaps compromised by lower DMFS compared to ¹⁰³Pd; though a mechanism for a DMFS advantage for ¹⁰³Pd remains unclear and most likely this finding is due to imbalances between patient groups. At present, both radioisotopes remain appropriate for this population, and it seems that a clear answer in terms of the superiority of one over the other will not be possible without prospective studies.

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Conflicts of interest

There are no conflicts of interest.

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